








Modelling Financial Markets

Chapter 17: Bubbles and Crashes



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Can Statistics Spot a Bubble Before It Bursts?

Bubbles, Crashes and Their Detection

- Prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value
- Questions
  - can statistics detect such episodes **in real time**?
  - can it say **when** they end?
- Part I: what a bubble is, run-ups and crashes in the data, tests for explosive prices and their date-stamping
- Part II: the LPPLS model (Log-Periodic Power Law Singularity), regime switching, crypto and artificial-intelligence (AI) stocks, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the Appendix at the end

What you should be able to do

- separate the fundamental value from a rational bubble and state what can be tested
- test for explosive prices on rolling windows and date-stamp the episodes, with valid critical values
- fit LPPLS, judge the stability of its critical time and evaluate its alarms honestly
- describe booms and busts with a Markov-switching model and test the number of regimes

Core Path, Applications and Proofs

- **Core path**: the main slides of four sections, in order
  - What Is a Bubble?, Testing for Explosive Behaviour, the LPPLS model, Regime-Switching Models; then Which Tool When?
  - Two hand-worked examples carry the core logic: why the test uses log prices, and how a run of exceedances becomes a dated episode
- **Applications**, read after the core
  - Run-ups and crashes in the data, crypto and AI stocks, the housing case study, the AI-assisted research loop
- ▸ Appendix, optional depth
  - Derivations, limit theory, estimation details and caveats; the blue buttons on the main slides lead there and back
- **Seminar 17**: numbers cited next to an exercise are replication targets
  - Reproduce the number first; if yours differs, compare the sample period, the minimum window and the critical values before the code

Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain; Wikimedia Commons

- **Tulips, 1636–1637** Garber (1990)
  - Contracts on rare bulbs traded at very high prices until the market stopped in February 1637
  - Part of the path fits the fundamentals of a new luxury good
  - Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720** Kindleberger & Aliber (2005)
  - Shares rose several-fold within months, then collapsed
  - Ingredients: a financial innovation, easy credit, a story about future profits
  - Lesson 2: bubbles come with credit and with high turnover Scheinkman & Xiong (2003)
- **1929** Kindleberger & Aliber (2005)
  - A boom partly financed with borrowed money ended in the October crash
  - Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic
  - Lesson 3: we need tests that work on a single time series, in real time, before the panic

The Series Tested: Stocks, Industries, Crypto

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2026	price-dividend ratio
48 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange indices
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy (formerly MicroStrategy)	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; Kenneth French Data Library for industries

Fundamental Value

- No-arbitrage pricing: $P_t = \mathbb{E}_t[M_{t+\mathbf{1}}(P_{t+\mathbf{1}} + D_{t+\mathbf{1}})]$
  - P_t: price at the end of period t; $D_{t+\mathbf{1}}$: dividend paid in period $t+1$
  - $M_{t+\mathbf{1}}$: stochastic discount factor, the random factor that converts a payoff at $t+1$ into its value at t; $\mathbb{E}_t$: expectation under the physical measure (the probabilities of actual outcomes), given the information at t
- **Model assumption**: a constant required (expected) return r
  - Then $P_t = \mathbb{E}_t[P_{t+\mathbf{1}} + D_{t+\mathbf{1}}]/(1+r)$: today's price is the discounted expected price plus dividend
  - In effect $M_{t+\mathbf{1}}$ is replaced by the constant $1/(1+r)$; the rest of the chapter uses only this case
- Solving forward gives the **fundamental value**: $F_t = \sum_{i \geq \mathbf{1}} \mathbb{E}_t[D_{t+i}]/(1+r)^i$
  - The present value of all expected dividends
  - Gordon case, dividends growing at $g < r$: $F_t = D_{t+\mathbf{1}}/(r-g)$
- But $P_t = F_t + B_t$ also solves the pricing equation
  - B_t: any extra term with $\mathbb{E}_t[B_{t+\mathbf{1}}] = (1+r)\,B_t$
  - B_t is the **rational bubble**: investors pay for it because they expect to resell at a higher price Blanchard & Watson (1982)

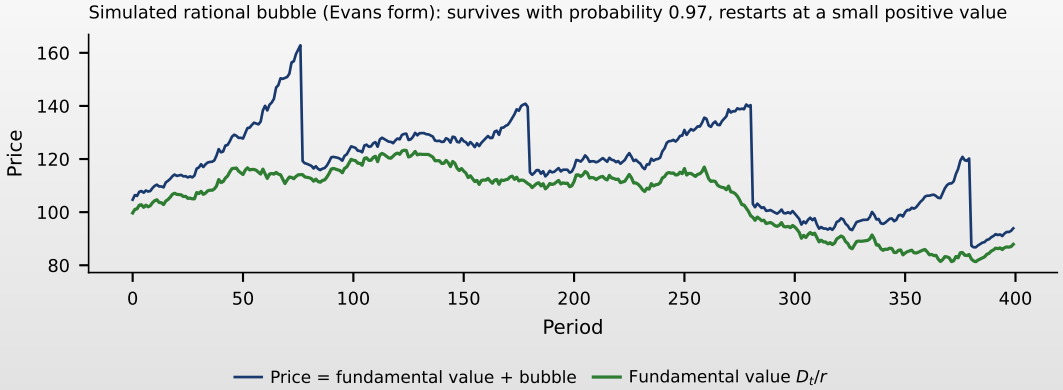
Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$ for any horizon of h periods, an **explosive conditional mean**; individual paths can crash
  - The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- Rational bubbles are positive and exist from the first trade Diba & Grossman (1988)
  - A negative bubble would eventually push the expected price below zero
  - A bubble that is zero today must stay zero
- A bubble is the failure of the **transversality condition** $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$
  - The price contains a value never paid out as dividends (derivation in the Appendix)
- **Testable implication**
  - If dividends are at most I(1), then without a bubble the price and the price-dividend ratio are **not explosive**
  - I(1): integrated of order one; the first difference is I(0) (stationary), while the level is not

Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1-\pi$ Blanchard & Watson (1982)
  - $B_{t+1} = \dfrac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π, $B_{t+1} = \varepsilon_{t+1}$ otherwise
  - $\pi \in (0,1)$: survival probability; ε_{t+1}: a shock with mean zero given the information at t
- Expected growth is still r; while it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time
- After a collapse $B = \varepsilon$ can be negative, which Diba–Grossman rule out: the simulation on the next slide restarts from a positive value (Evans form) Evans (1991)

A Simulated Rational Bubble (1/2)



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A Simulated Rational Bubble (2/2)

- Fundamental value D_t/r, random-walk dividends; $r = 1\%$, $\pi = 0.97$
  - While alive and large the bubble grows by about 4.1% per period; expected life 33.3 periods
- Evans form: after a collapse the bubble restarts from $\delta = 5$ Evans (1991)
  - Survival (probability π): $B_{t+\mathbf{1}} = \left[\delta + \frac{\mathbf{1}+r}{\pi}(B_t - \frac{\delta}{\mathbf{1}+r})\right] u_{t+\mathbf{1}}$; collapse (probability $1-\pi$): $B_{t+\mathbf{1}} = \delta\, u_{t+\mathbf{1}}$
  - $u_{t+\mathbf{1}}$: noise with mean 1, drawn independently of survival; hence $\mathbb{E}_t[B_{t+\mathbf{1}}] = \pi\left[\delta + \frac{\mathbf{1}+r}{\pi} B_t - \frac{\delta}{\pi}\right] + (1-\pi)\delta = (1+r)B_t$
  - The simulation uses these two branches at every B_t, without the small-bubble threshold of Evans' original form
- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 1.43 times the fundamental value
- Between episodes the price tracks fundamentals
  - A full-sample test sees mostly "normal" data: the tests of this chapter use **windows**

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Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017); CC BY 2.0; Wikimedia Commons

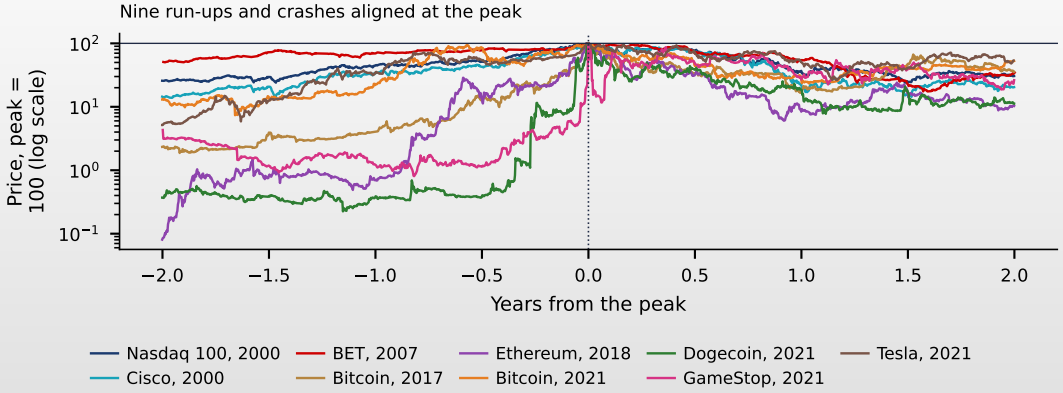
- **Feedback and narratives** Shiller (2015)
  - Rising prices attract new buyers
  - "New era" stories spread by word of mouth
- **Disagreement and short-sale limits** Scheinkman & Xiong (2003)
  - The price includes the option to resell to a more optimistic buyer
- **Riding the bubble** Abreu & Brunnermeier (2003), Brunnermeier & Nagel (2004)
  - Informed traders stay in as long as others do, since no one can burst it alone
- Common signature of all three
  - Prices grow faster than fundamentals for a while: the explosive pattern we will test

Nine Episodes: Run-Up and Crash

Episode	Peak	2-year run-up	Max. fall	Trough	Back at peak
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4 186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124 004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26 897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2 218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1 837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode; run-up: simple percentage change of the price over the two years before the peak, $P_{\mathbf{peak}}/P_{\mathbf{peak}-\mathbf{2y}} - 1$; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak (1/2)



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Run-Ups and Crashes Aligned at the Peak (2/2)

- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ enormously, from +97% (BET) to +124 004% (Ethereum); the falls are much more alike, 74–94%
- Common shape: an accelerating rise, a sharp top, a deep fall
  - Convex on the log scale: the growth rate itself rises, the **explosive** behaviour the tests look for
- The run-ups in the table are simple percentage changes; the tests use log returns $\ln(P_t/P_{t-1})$
  - Example: a doubling is +100% as a simple change, but $\ln 2 \approx 0.69$ as a log return
  - Log returns add up over time and match the log scale of the chart; simple changes compound
- Selection bias: we picked these episodes because they crashed
  - Many run-ups were not followed by crashes

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The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia Commons

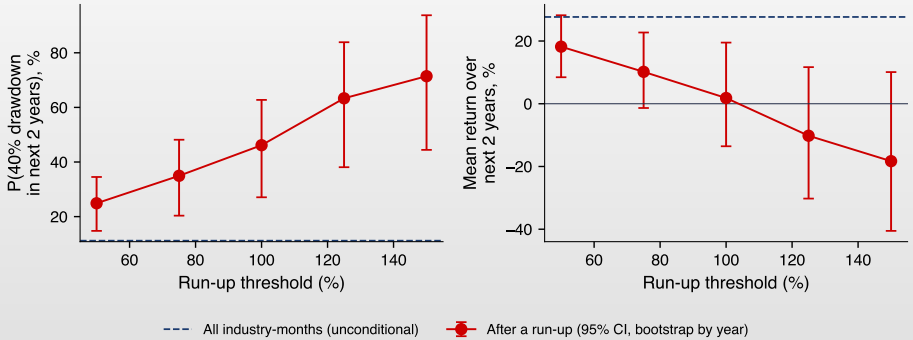
- After the peak of 27 Mar 2000: Nasdaq 100 −83%, Cisco −89% (table)
- Hedge funds rode the bubble and cut positions before the fall Brunnermeier & Nagel (2004)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble Pástor & Veronesi (2009)

Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news Fama (2014)
- Greenwood, Shleifer and You answer with **all** US industries since 1926, not with famous episodes only: do sharp run-ups predict crashes, or low returns? Greenwood, Shleifer & You (2019)
- **Run-up**: value-weighted industry return (each firm weighted by its market capitalisation) over the past 2 years above a threshold (e.g. 100%)
  - Both conditions at once: the raw return **and** the return net of the market exceed the threshold
  - And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted (Section 2 of the paper)
  - First month of the run-up; a new episode in the same industry only two years later
- **Crash**
  - A drawdown of at least 40% from a previous high within the next 2 years
  - The next chart gives each crash frequency with a 95% CI (Confidence Interval)
- Our replication, same sample rules: the first 48 of the 49 industries of the Kenneth French Data Library (without "Other"), months with at least ten firms: 41 679 industry-months, 1926–2026

What Do Sharp Run-Ups Predict? (1/2)

Bubbles for Fama, 48 US industries 1926-2026: two-year run-ups above the threshold, raw and net of the market



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What Do Sharp Run-Ups Predict? (2/2)

- After a run-up above 100% (52 episodes in 27 industries), a 40% crash followed in 46% of cases, 95% CI [27%, 63%]
  - Run-up above 100% both raw and net of the market; the five chart points are the thresholds 50, 75, 100, 125 and 150%
  - Against 11% for a random industry-month; above 150% (rightmost point): 71% ($n = 21$)
  - CI: bootstrap over calendar years, because run-ups cluster in time (e.g. 1928–1929, 1967–1968, 1980); effective sample size in the Appendix
- The mean return over the next 2 years is not significantly different from zero
  - 2% after a 100% run-up, CI $[-14\%, 20\%]$: no statistically established mean loss
- Run-ups raise the **risk** of a crash, not the predictable loss: price run-ups alone do not give a profitable timing signal Greenwood, Shleifer & You (2019)

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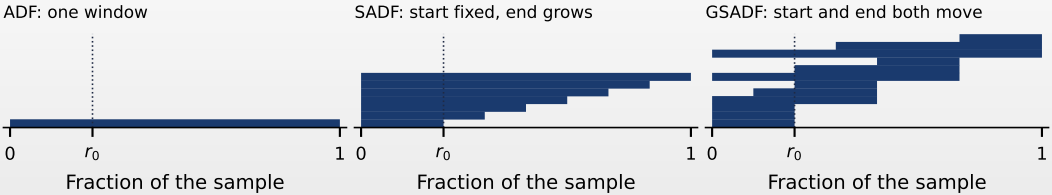
Right-Tailed ADF and Collapsing Bubbles

- Regression on the log price y_t: $\Delta y_t = a + b\, y_{t-1} + \varepsilon_t$
  - $\Delta y_t = y_t - y_{t-1} = \ln(P_t/P_{t-1})$: the log return; a: constant; ε_t: shock; here no lagged Δy
  - $\mathrm{ADF} = \hat{b}/\mathrm{SE}(\hat{b})$: the Augmented Dickey–Fuller (ADF) t-statistic, called t_b in Seminar 17; $\hat{b}$: the OLS (Ordinary Least Squares) estimate; SE: its Standard Error
  - $b < 0$: mean reversion; $b = 0$: random walk (unit root); $b > 0$: each rise feeds the next, an explosive path
- Bubble test: $H_0 : b = 0$ (unit root) vs $H_1 : b > 0$ (explosive root), the **right tail** Phillips, Wu & Yu (2011)
  - The left-tail Dickey–Fuller table does not apply
- Evans: a bubble that collapses periodically leaves the **full-sample** test with low power Evans (1991)
  - The whole series looks stationary or like a random walk
  - Collapses give the full-sample slope a negative sign (Appendix)
- Solution: run the ADF regression on many **windows** and take the largest statistic
  - SADF (Sup Augmented Dickey–Fuller) Phillips, Wu & Yu (2011) and GSADF (Generalised Sup Augmented Dickey–Fuller) Phillips, Shi & Yu (2015a) differ in the windows they search (next two slides)

Levels Explode, Log Prices Accelerate

- Theory: the bubble is explosive in **levels**, $\mathbb{E}_t[B_{t+\mathbf{1}}] = (1+r)B_t$; the test regresses the **log** price
  - A pure bubble path $B_t = B_{\mathbf{0}}(1+r)^t$ has a constant log growth $\ln(1+r)$: Δy_t does not depend on $y_{t-\mathbf{1}}$, so $b = 0$
  - What $b > 0$ detects: a **rising** growth rate of $\ln P_t$, which occurs while the bubble share of $P_t = F_t + B_t$ grows
- Toy example ($1 + r = 2$ for easy arithmetic): $F = 100$ constant, $B = 1, 2, 4, 8$, so $P = 101, 102, 104, 108$
  - Bubble share of the price: 1.0%, 2.0%, 3.8%, 7.4%
  - Log increments: $\ln(102/101) = 0.0099$, $\ln(104/102) = 0.0194$, $\ln(108/104) = 0.0377$
  - They rise with $y_{t-\mathbf{1}} = \ln P_{t-\mathbf{1}}$, so the fitted $b > 0$; the bubble alone, $B_t = 2^t$, has every increment equal to $\ln 2 = 0.693$, so $b = 0$
- Check: add $B = 16$, so $P = 116$; is the next increment larger?
  - Yes, $\ln(116/108) = 0.0715$; once $B \gg F$ the increments approach $\ln 2$ and b falls back towards 0: windows that catch the **transition** have the power
- Derivation and the role of the deterministic terms: Explosive in Levels, Accelerating in Logs

Three Ways to Choose Windows (1/2)



- $\mathrm{ADF}_{r_1}^{r_2}$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample
  - Each bar in the chart is one ADF regression
  - r_0: the minimum window, as a fraction of the T changes Δy_t (observations minus one); sup: the largest value over the windows allowed
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$ Phillips, Shi & Yu (2015a)
  - E.g. $T = 440$ months: $r_0 = 0.096$, i.e. 42 months

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Three Ways to Choose Windows (2/2)

- SADF Phillips, Wu & Yu (2011)
  - $\mathrm{SADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}$
  - Windows start at the first observation and end anywhere after r_0
- GSADF Phillips, Shi & Yu (2015a)
  - $\mathrm{GSADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$
  - Windows start and end anywhere, with length at least r_0
- BSADF (Backward Sup Augmented Dickey–Fuller): backward SADF at the end point r_2 Phillips, Shi & Yu (2015a)
  - $\mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$, $\mathrm{GSADF} = \sup_{r_2} \mathrm{BSADF}_{r_2}$
  - BSADF_{r_2} uses only data up to r_2: it can be computed **in real time**
- Larger search, higher bar
  - The more windows a statistic searches, the larger its critical value: simulated for each T and r_0

Date-Stamping an Episode (1/2): the PSY Rules

- **Origination**, in the PSY (Phillips, Shi and Yu) procedure Phillips, Shi & Yu (2015a)
  - The first date at which BSADF_{r_2} exceeds its critical value cv_{r_2}
  - cv_{r_2}: the 95% quantile of BSADF_{r_2} under the null, date by date
- **Termination** (PSY, eq. 8) Phillips, Shi & Yu (2015a)
  - The first date after $\hat{r}_e + \delta \ln(T)/T$ at which BSADF falls below cv_{r_2}
  - $\hat{r}_e$: the estimated start, as a fraction of the sample; T: number of changes Δy_t
  - $\delta \ln(T)/T$: a minimum duration, with δ chosen for the data frequency
- PWY (Phillips, Wu and Yu) date-stamping uses the recursive $\mathrm{ADF}_0^{r_2}$ instead Phillips, Shi & Yu (2015b)
  - It may miss a second bubble after a collapse

Date-Stamping an Episode (2/2): the Rule Used Here

- Minimum-duration rule of the R package exuber Vasilopoulos, Pavlidis & Martínez-García (2022)
  - Runs of consecutive exceedances shorter than L periods are dropped
  - $L = \lceil \ln T \rceil$, i.e. $\delta = 1$; $\lceil \cdot \rceil$: rounded up; $T = 440$ months gives $L = 7$
- Start and end of an episode
  - Start: the first exceedance; end: the last exceedance (exuber dates the end one period later)
  - Eq. 8 would instead extend an isolated one-date exceedance to L dates
- Real time
  - The start is dated in hindsight
  - The episode is **confirmed** only after L consecutive exceedances, $L - 1$ periods after its start
- **Chain**: kept runs separated by a short gap, used for the leads in Real Time vs Hindsight
  - Monthly data: the next run starts at most 93 days after the previous one ends (at most two months without an exceedance)

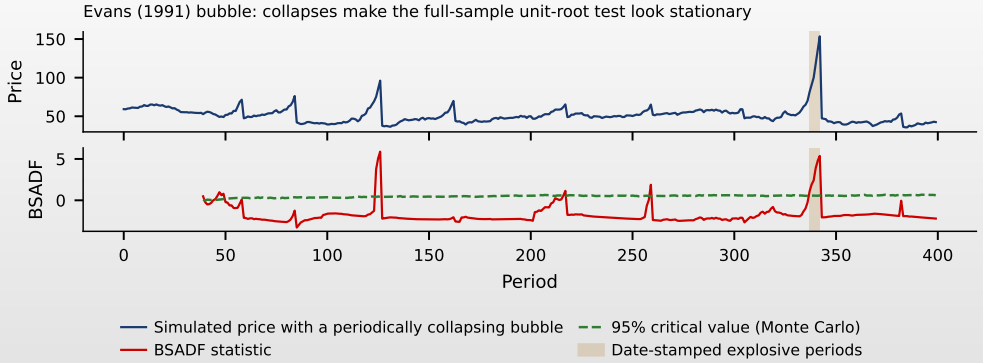
A Hand-Worked Date-Stamping Example

- Invented numbers; $L = 3$; critical value $cv_t = 1.0$ at every date; flag $= 1$ if $\mathrm{BSADF}_t > cv_t$; s: start, c: confirmed, e: end, d: dropped

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
BSADF_t	0.4	1.2	1.5	1.8	1.3	0.8	1.1	1.4	1.9	1.6	0.7	0.5	1.3	1.2	0.6
Flag	0	1	1	1	1	0	1	1	1	1	0	0	1	1	0
Event		s		c	e		s		c	e			d	d	

- Runs $t = 2$–5 and $t = 7$–10: four exceedances each, at least L, so both are kept
  - Confirmed at $t = 2 + L - 1 = 4$ and $t = 7 + L - 1 = 9$; each ends at its last exceedance, $t = 5$ and $t = 10$
  - Run $t = 13$–14: only two exceedances, fewer than L: dropped, never confirmed
- A toy chain allowing at most one quiet date between runs links 2–5 and 7–10 (one quiet date, $t = 6$). The monthly rule allows up to two quiet months (93 days)
  - Episode containing $t = 10$: strict rule start $t = 7$; chain start $t = 2$, so the measured lead grows by five periods
- Check: flags 1 1 0 1 1 1 1 with $L = 3$; when is the first episode confirmed? Answer: at $t = 6$; the run at $t = 1$–2 is too short

Evans Bubble: Full Sample vs Windows (1/2)



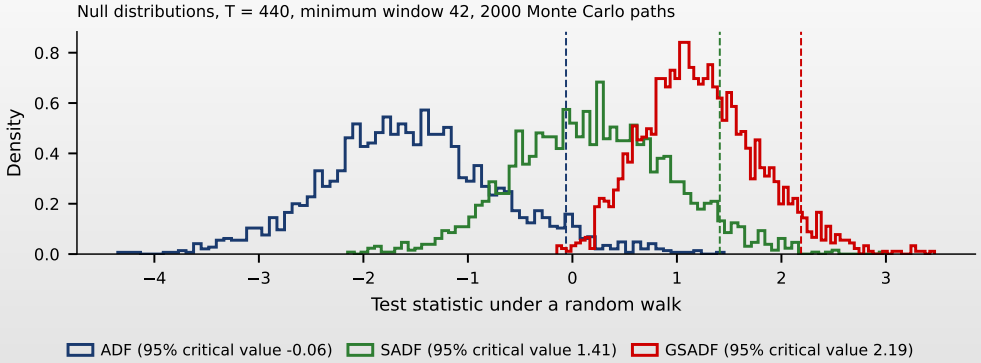
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Evans Bubble: Full Sample vs Windows (2/2)

- 400 simulated periods; BSADF against its 95% critical value from 2 000 random-walk paths
- Full-sample right-tailed ADF: -5.49 vs 95% critical value -0.09: no rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46) and GSADF 5.88 (critical value 2.21)
- BSADF crosses its critical value in several bursts, but only one lasts long enough to be date-stamped
  - The minimum-duration rule ignores short bursts and so reduces false alarms

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Null Distributions and Critical Values (1/2)



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Null Distributions and Critical Values (2/2)

- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2 000 paths of $T = 440$; 95% critical values: a single ADF -0.06 (negative!), SADF 1.41, GSADF 2.19 Phillips, Shi & Yu (2015a)
- A supremum over more windows shifts the distribution to the right: the GSADF bar is the price of searching over about 80 000 windows; simulate it for each T, r_0 and lag order
- Asymptotics (▸ Appendix): the null limit is pivotal; against mildly explosive roots the tests are consistent; date-stamping keeps a 5% false-alarm chance at every date Phillips, Shi & Yu (2015b)

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Changing Volatility: Wild Bootstrap

- Monte Carlo critical values assume shocks with constant variance
  - Real prices show volatility clustering and shifts in volatility
  - A burst of volatility can mimic explosiveness and inflate the rejection rate Harvey et al. (2016)
- **Wild bootstrap** Phillips & Shi (2020)
  - Multiply the observed, demeaned changes Δy_t by random signs $w_t = \pm 1$, rebuild the path, recompute BSADF
  - The bootstrap data have a unit root by construction (no explosive root) and inherit the time pattern of $\mathrm{Var}(\Delta y_t)$ Harvey et al. (2016)
  - Demeaning removes the drift, as in the null model
- Retrospective and real-time versions
  - Centring with the full-sample mean and resampling the whole path give **retrospective** critical values
  - In real time, recompute them with data up to each date

Size of GSADF When Volatility Changes: Design

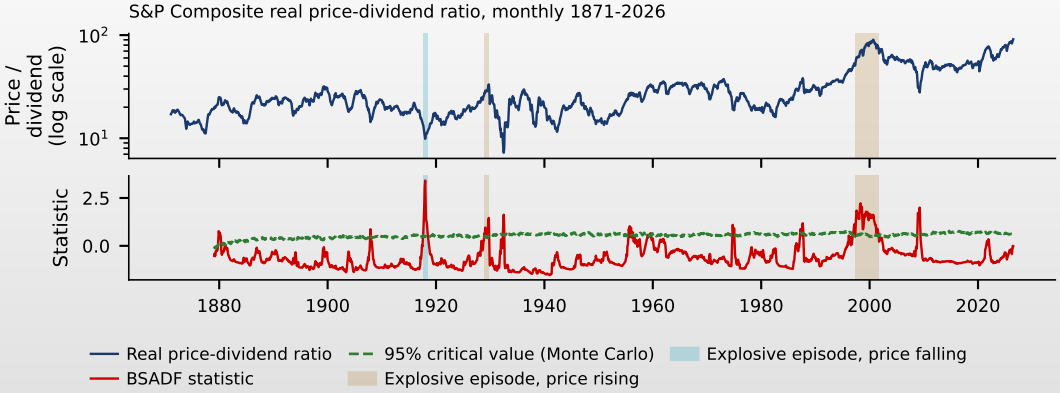
- Null model without a bubble: $y_t = T^{-1} + y_{t-1} + e_t$, a random walk with a small drift
  - e_t: shocks with mean zero and the variance pattern of each scenario
- Four variance patterns
  - Constant; multiplied by 3 at $T/2$; divided by 3 at $T/2$
  - GARCH (Generalised AutoRegressive Conditional Heteroskedasticity)(1,1): $\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$, with $\alpha = 0.10$, $\beta = 0.88$
  - σ_t^2: conditional variance; $\omega > 0$: constant; α: reaction to the last shock; β: persistence
- Sample: $T = 400$, minimum window $w_0 = \lfloor r_0 T \rfloor = 40$ observations
  - 1 000 paths per scenario, each with its own wild bootstrap (199 replications)
  - Monte Carlo error: ± 1.4 percentage points near 5%
- Two outcomes per path
  - Size: GSADF above its 95% critical value (Monte Carlo, MC in the table, or wild bootstrap)
  - False episode: BSADF above its pointwise 95% value for 6 consecutive periods

Size of GSADF When Volatility Changes: Results

Null model, no bubble	GSADF, MC crit. value	GSADF, wild crit. value	False episode, MC	False episode, wild
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%
Volatility $\times 1/3$ at $T/2$	4.3%	6.1%	48.0%	54.8%
GARCH(1,1), $\alpha = 0.10$, $\beta = 0.88$	16.6%	6.8%	58.1%	52.8%

- A late rise in volatility is mistaken for explosiveness Harvey et al. (2016)
  - Size 32.0% with Monte Carlo, 4.1% with wild bootstrap critical values
  - Under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless
- Even with constant volatility, 49.1% of the paths contain a false date-stamped episode
  - The size is controlled date by date, not over the whole sample
- The GARCH and $\times 1/3$ rows are the replication target of Seminar 17, B9

S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (1/2)



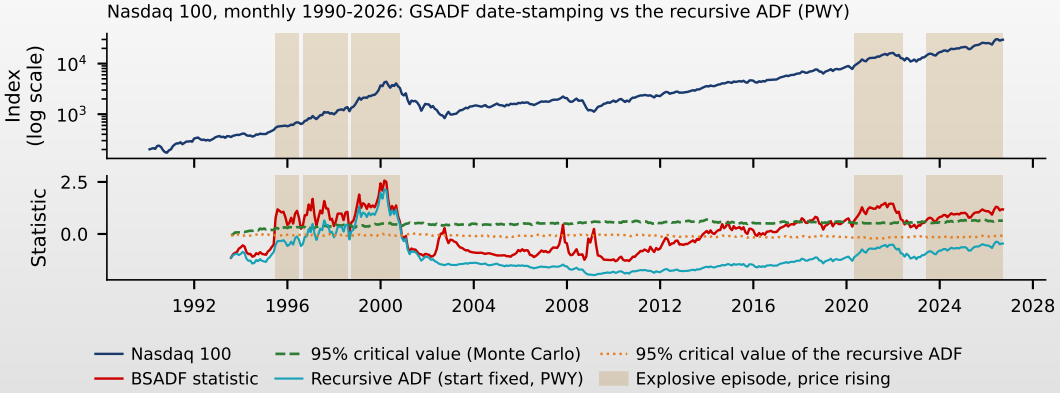
MFM_ch17_sp_pd

S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (2/2)

- Real price over real dividend, monthly, Jan 1871–Jun 2026
  - $T = 1\,865$, minimum window 96 months
  - Shiller's S&P Composite, not the S&P 500 series; in the chart legend, "price rising/falling" refers to the plotted series, here the ratio
- GSADF $= 3.41$ vs 95% critical value 2.39 (99%: 2.88)
  - The full-sample ADF (-1.11) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
  - 1917–1918: an accelerating **fall** of the ratio, explosive but not a bubble
- Dot-com: peak 90.2 in Aug 2000, long-run mean 29.8
  - The sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not necessarily explosive
- A ratio can explode through its denominator: check both price and dividend

MFM_ch17_sp_pd

Nasdaq 100: GSADF vs PWY Date-Stamping (1/2)



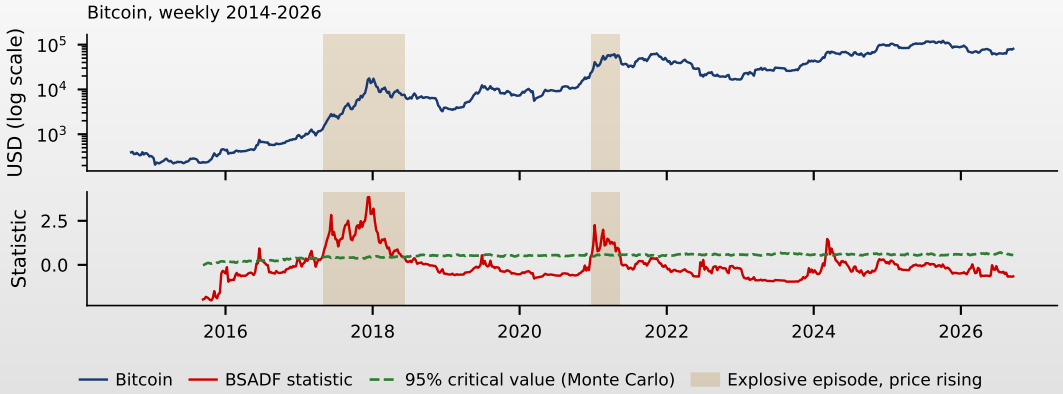
MFM_ch17_nasdaq

Nasdaq 100: GSADF vs PWY Date-Stamping (2/2)

- Monthly log closes 1990–2026, $T = 440$
  - Minimum window 42 months; 2 000 Monte Carlo paths
- GSADF $= 2.57$ (95% critical value 2.19); SADF $= 2.14$ (1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
  - PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF misses the post-2020 episodes
  - It starts in 1990 and is dragged down by the long flat period after 2000
- Dot-com episode dated from Jun 1995: the start of a chain of 3 BSADF runs
  - 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995
  - The run that contains the peak alone starts in Sep 1998

MFM_ch17_nasdaq

Bitcoin: Explosive Episodes (1/2)



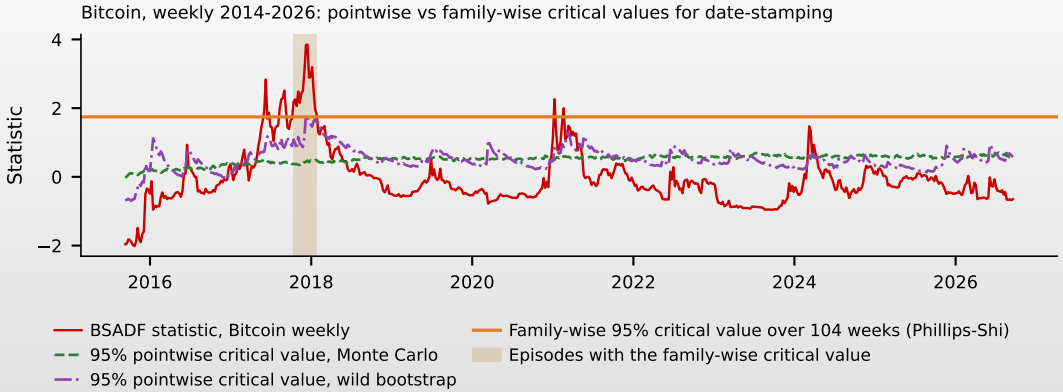
MFM_ch17_bitcoin

Bitcoin: Explosive Episodes (2/2)

- Weekly log closes in US dollars (USD) 2014–2026, $T = 626$
  - Minimum window 51 weeks; Monte Carlo critical values
- GSADF $= 3.85$: rejected with either critical value
  - 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
  - Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: episode dated from Apr 2017
  - 8 months before the peak of 16 Dec 2017; confirmed in real time in Jun 2017
  - It ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged
  - Steady growth is not explosive acceleration

MFM_ch17_bitcoin

Date-Stamping Is a Multiple Test: Bitcoin (1/2)



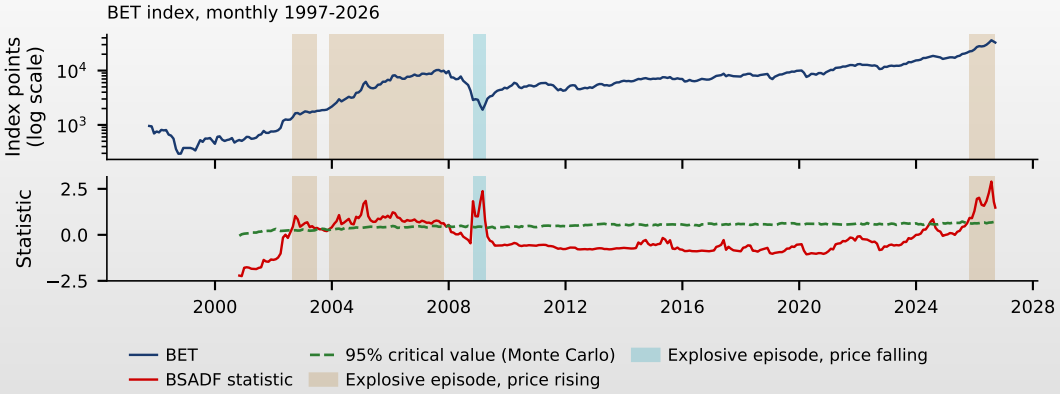
MFM_ch17_inference

Date-Stamping Is a Multiple Test: Bitcoin (2/2)

- BSADF vs its critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER)
  - The probability of at least one false alarm over the whole monitoring window
  - 2 000 random walks with the Bitcoin T and w_0: a crossing in 95%, an "episode" in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF Phillips & Shi (2020)
  - Over the last τ_b monitoring dates; here $\tau_b = 104$ weeks
  - Family-wise critical value 1.75, against Monte Carlo pointwise values of at most 0.71; wild-bootstrap pointwise values can be higher because they retain time-varying volatility, unlike this i.i.d. residual bootstrap
- Episodes above the family-wise threshold: Oct 2017–Jan 2018
  - The 2020–2021 episode found with Monte Carlo pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the Appendix)

MFM_ch17_inference

BET: The 2007 Boom and 2025 (1/2)



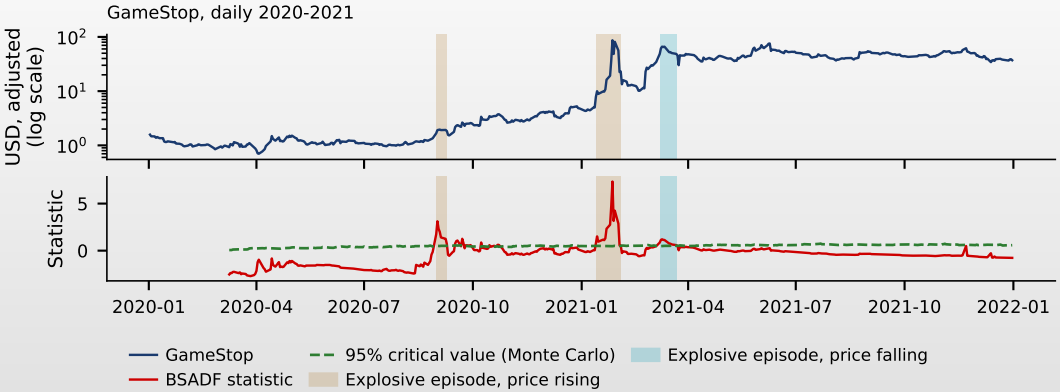
MFM_ch17_bet_gamestop

BET: The 2007 Boom and 2025 (2/2)

- Monthly log closes of the official BET, 1997–2026
  - $T = 348$, minimum window 37 months
- GSADF $= 2.90$ vs 95% critical value 2.21
  - Episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up around Romania's accession to the European Union (January 2007)
  - BET then fell 83% from its peak of 24 Jul 2007
- Oct 2008–Mar 2009: an accelerating **fall**
  - Explosive can mean down as well as up
- The current episode (from Oct 2025) is still open
  - Explosive does not mean it will crash: the rise is accelerating

MFM_ch17_bet_gamestop

GameStop, January 2021 (1/2)



MFM_ch17_bet_gamestop

GameStop, January 2021 (2/2)



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- Daily log closes 2020–2021, $T = 504$, window 45 days; GSADF $= 7.34$, critical value 2.21
- January episode dated from 13 Jan 2021, 14 days before the closing high of 27 Jan 2021; confirmed on 22 Jan 2021
- A surge of a few weeks: detecting it needs daily data
  - The test does not identify the mechanism (short covering: buying back shares sold short)

MFM_ch17_bet_gamestop

Real Time vs Hindsight

Episode	Start / Peak	Confirmed	Lead (months)	Lead from conf.	Rise after conf.	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	Dec 1995	57	51	+716%	−76%
BET, monthly	Nov 2003 / Jul 2007	Apr 2004	44	39	+299%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	Jun 2017	8	6	+590%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	Feb 2021	4	2	+66%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	22 Jan 2021	0.5	0.2	+435%	−88%

- Columns (caveats in the ▸ Appendix)
  - Start: first exceedance of the chain of BSADF runs that leads to the peak (hindsight; chain defined on the next slide); confirmed: after L consecutive exceedances (real time)
  - Leads in months; rise: from confirmation to the peak; fall: to the lowest close within 2 years
- Even at confirmation the tests **warn early**, but they do not time the crash Phillips, Shi & Yu (2015b)
  - The lead ranges from 5 days to about 4.2 years
  - Selling at confirmation would have missed most of the rise

Reading the Real-Time Table

- **Chain**: date-stamped runs joined when the next starts soon after the previous ends
  - At most 93 calendar days later for monthly data (two months without an exceedance), 28 days for weekly data, 10 days for daily GameStop
  - A hand-worked case: A Hand-Worked Date-Stamping Example
- Nasdaq 100: the chain joins 3 runs, from Jun 1995; the strict rule keeps only the run containing the peak
  - Strict rule: start Sep 1998, confirmed Mar 1999, leads 18 and 12 months instead of 57 and 51
  - For BET, Bitcoin and GameStop the two rules give the same start
- Two falls for the same dot-com crash
  - This table: -76%, the lowest daily close within 2 years of the peak
  - Nine Episodes table: -83% within 4 years; the trough of Oct 2002 came 30 months after the peak, outside the 2-year window
- Bitcoin 2021: the peak here is the local high of 13 Apr 2021, not the all-time high of 8 Nov 2021
  - The explosive chain ended in May 2021; the later high came after a deep fall and a slower recovery that BSADF did not flag

Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0 (Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS) Johansen, Ledoit & Sornette (2000)
  - Imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches
  - Hazard rate: the crash probability per unit time, given no crash yet
- Rational investors stay in only if the price grows fast enough
  - The extra growth pays for the rising crash risk
- Result: **faster-than-exponential** growth
  - Decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date Sornette (2017)

The LPPLS Equation

- $\ln p(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
  - In words: the log price is a power-law trend towards the critical time plus oscillations whose frequency rises as t_c approaches; $p(t)$: price at time t; t_c: critical time
- A: log price at t_c; $B < 0$: the super-exponential trend for a rising bubble; $0 < m < 1$: power-law exponent
  - The growth rate of the power-law trend, $-Bm(t_c - t)^{m-1}$, rises without bound as $t \to t_c$
- C, ω, ϕ: amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at t_c

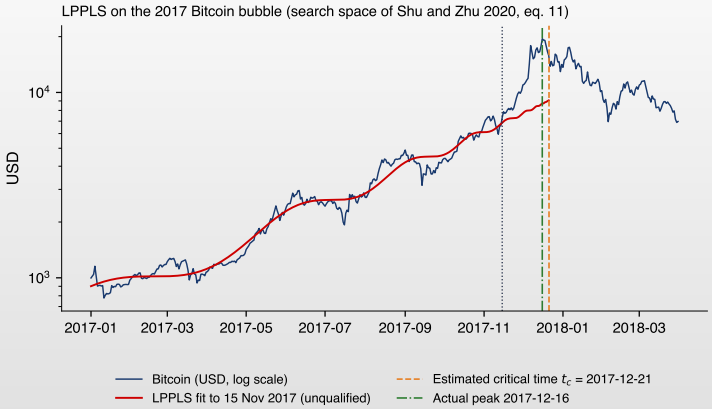
Calibration: Filimonov and Sornette

- Rewrite the cosine: $C\cos(\omega\ln(t_c - t) - \phi) = C_1\cos(\omega\ln(t_c - t)) + C_2\sin(\omega\ln(t_c - t))$ Filimonov & Sornette (2013)
  - $C_1 = C\cos\phi$, $C_2 = C\sin\phi$; A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS on the window $[t_1, t_2]$
- Search only over (t_c, m, ω): a grid, then a Nelder–Mead refinement (a derivative-free simplex search) of the sum of squared residuals
- Search space of Shu and Zhu (Section 2.3, eq. 11) Shu & Zhu (2020)
  - $m \in [0, 1]$, $\omega \in [1, 50]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$
  - Damping $\frac{m|B|}{\omega|C|} \geq 1$: an approximate condition for a non-negative crash hazard rate
  - Exact condition: $m|B| \geq |C|\sqrt{m^2 + \omega^2}$ (Seminar 17, A8)
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Qualified Fits: The Filter

- A fit on $[t_1, t_2]$ is **qualified** if it passes all conditions of Shu and Zhu (Section 2.3, eq. 12), and $B < 0$ for a rising bubble Shu & Zhu (2020)
  - $m \in [0.01, 0.99]$, $\omega \in [2, 25]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
  - Damping $D = m|B|/(\omega|C|) \geq 1$: the search constraint is also required by the qualification filter here
  - Number of oscillations (half-periods) $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not noise
  - Largest relative error of the fitted price $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
  - Lomb periodogram of the detrended residual $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$: log-periodicity significant at 10%
  - Lomb periodogram: spectral power at each frequency for unevenly spaced points, here the values of $\ln(t_c - t)$; significant: a spectral peak that noise alone would rarely produce
  - Residual unit-root check for $\ln \hat{p}_t - \ln p_t$: Dickey–Fuller and Phillips–Perron reject a unit root at 10% (a stationary, mean-reverting residual; this alone does not make it an autoregressive AR(1) process)
  - Phillips–Perron: a Dickey–Fuller-type unit-root test with a nonparametric correction for serial correlation
- $[t_1, t_2]$: the estimation window; the filter is applied to each window separately

LPPLS on Bitcoin, 2017 (1/2)



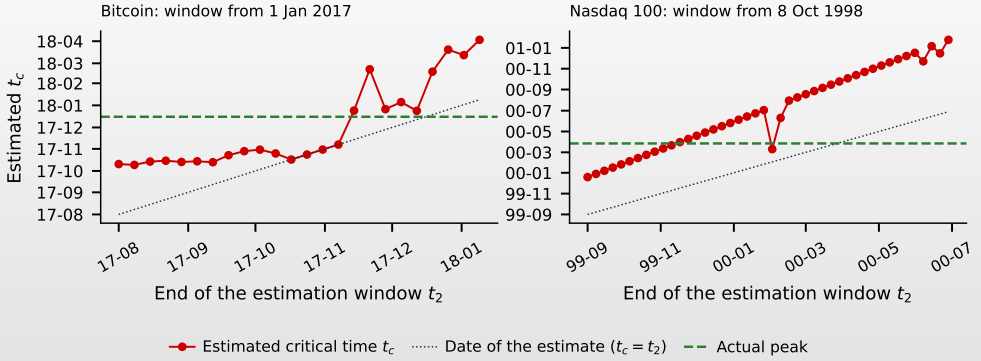
MFM_ch17_lppls

LPPLS on Bitcoin, 2017 (2/2)

- Fit on daily log prices, 1 January to 15 November 2017
  - The path after 15 November is out of sample
- The fit follows the 2017 rise and its oscillations closely
  - $m = 0.95$, $\omega = 10.54$, $B = -2.62$
- Estimated t_c: 21 Dec 2017; actual peak: 16 Dec 2017
  - The peak came 5 days earlier
- The fit fails the condition on relative error ≤ 0.15
  - It does **not** pass the filter, even though it looks convincing

MFM_ch17_lppls

How Stable Is the Critical Time? (1/2)



MFM_ch17_lppls

How Stable Is the Critical Time? (2/2)

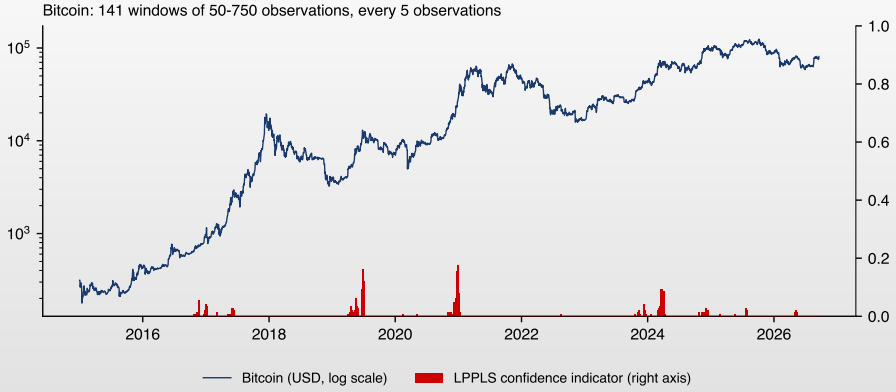
- A good in-sample fit is not a forecast
  - Each point: one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits) t_c ranged from Oct 2017 to Feb 2018
  - Mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Dec 1999 to Sep 2000
  - Mean absolute error 70 days; $t_c \geq t_2$ is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin), 0% (Nasdaq 100)
  - The birth of a bubble is easier to diagnose than its burst Demos & Sornette (2017)

MFM_ch17_lppls

The LPPLS Confidence Indicator

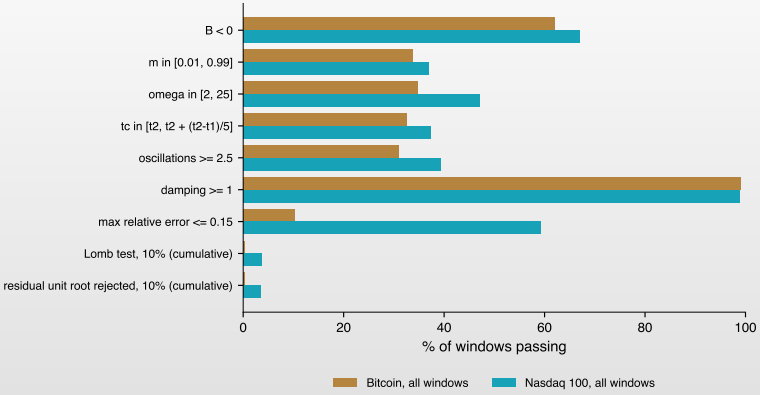
- At each date t_2, fit many windows $[t_2 - L, t_2]$ and report the **share of qualified fits** Demirer et al. (2019)
  - Grid of Shu and Zhu (Section 2.3): 141 windows, L from 750 down to 50 observations in steps of 5; t_2 every 5 observations; data up to t_2 only Shu & Zhu (2020)
  - Daily closes; Bitcoin trades every day, so one observation is one calendar day
- A high share means the bubble signature is robust to the choice of window
- Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
  - Fall measured from the price on the signal date; only dates whose 90 calendar days are fully observed (the last three months of data are dropped)

Bitcoin: LPPLS Confidence Indicator



MFM_ch17_lppls_confidence

LPPLS: Qualification Filters



MFM_ch17_lppls_confidence

Bitcoin: Real-Time Confidence Indicator

- Price panel: Bitcoin price and, as bars on the right axis, the share of the 141 windows with a qualified fit, every 5 days, 2016–2026
  - Filter panel: share of all windows passing each condition, Bitcoin and, for comparison, the Nasdaq 100
  - Cumulative rows: the Lomb and residual tests count only windows that pass every condition above them
- Positive on 14% of the 2017 dates, 3% in 2021, 12% in other years
  - Maximum 0.18 in Dec 2020
- A 20% fall within 90 days followed 22% of signals, 37% of no-signal dates
  - Bitcoin: 709 dates; Nasdaq 100, 1997–2026: 4% vs 11% (1482 dates)
- The raw difference is not yet evidence (test on the next slide)
  - Signals come in runs and the 90-day outcome windows overlap

MFM_ch17_lppls_confidence

Is the Indicator Better than Chance? A Test (1/2)

- H_0: $P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$
  - Statistic $\Delta = \hat{p}_{\text{on}} - \hat{p}_{\text{off}}$: share of signal dates followed by such a fall minus the share of no-signal dates
- Overlap: dates 5 days apart share up to 85 of their 90 outcome days
  - The two-sample z-test assumes independent dates and so understates the standard error
- Dependence-robust inference
  - Circular block bootstrap of the (signal, outcome) pairs: blocks of 90 days, 9 999 replications
  - Circular block bootstrap: resample blocks of consecutive dates, wrapping from the end of the sample to its start, so the dependence inside a block is kept
  - Linear probability regression of the outcome on the signal, with Newey–West standard errors
  - Newey–West: a standard error robust to autocorrelation up to a chosen lag, here 18 dates (90 days at 5-day spacing)

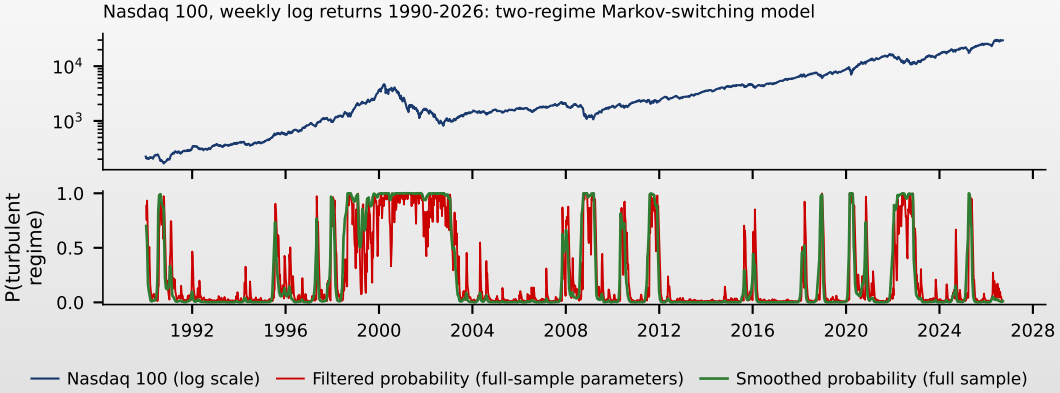
Is the Indicator Better than Chance? A Test (2/2)

- Bitcoin: $\Delta = -15.5$ points, 95% CI $[-30.0, 1.8]$; Newey–West two-sided $p = 0.049$ for $H_0 : \Delta = 0$
  - Block SE 8.1 vs independent-data SE 5.0
  - The percentile CI and the Newey–West t-test use different standard errors, so they need not agree near the 5% boundary
  - The two-sided p counts deviations in both directions; one-sided for a **lower** rate after signals, $p = 0.024$, a direction not stated in advance
  - The same test for the Nasdaq 100, with a power calculation: Seminar 17, Exercise B7
- Weak evidence: the block-bootstrap CI contains zero; the Newey–West regression is borderline (two-sided $p = 0.049$)
  - The point estimate is negative: if anything, a lower crash rate after signals
  - Signals come in steady rises, not just before crashes

The Markov-Switching Model

- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t}\varepsilon_t$, $\varepsilon_t \sim N(0,1)$, regime $s_t \in \{1,2\}$ unobserved Hamilton (1989)
  - $r_t = 100\ln(P_t/P_{t-1})$ in %, a log return, not the simple change $100(P_t/P_{t-1}-1)$
  - Each regime has its own mean μ and volatility σ: a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$; expected duration of regime i: $1/(1-p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates $P(s_t = j \mid r_1, \ldots, r_t)$ week by week
- **Filtered** probability: returns up to t for given parameters; **smoothed**: the whole sample (hindsight)
  - Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to t
  - A worked filter step is in Seminar 17, Exercise A2

Nasdaq 100: Two Regimes (1/2)



MFM_ch17_markov_switching

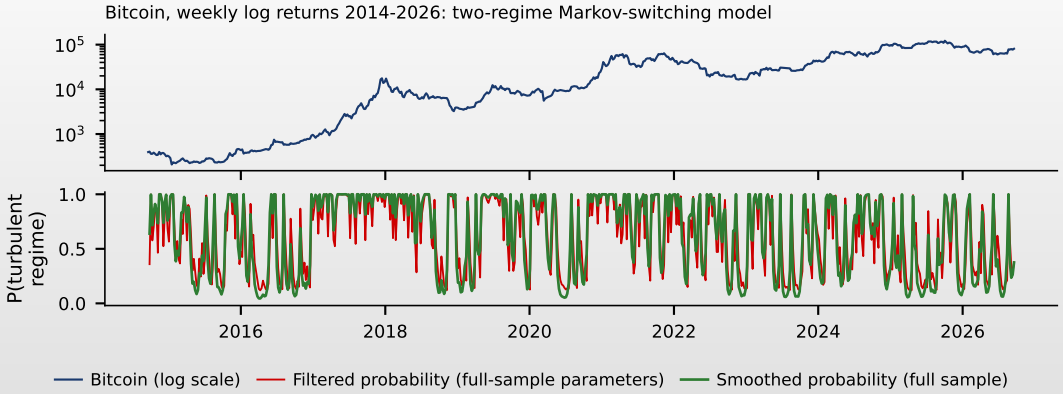
Nasdaq 100: Two Regimes (2/2)

Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{p}_{ii}$	Dur.
Calm	0.43 (0.06)	2.30	0.987	75.4
Turbulent	−0.32 (0.27)	5.40	0.957	23.5

- Weekly log returns in %, 1990–2026, 1 915 weeks
  - Duration in weeks; calm regime in 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5)
  - Filtered probability 0.99 at the end of March 2000: prices still rising, volatility jumping

MFM_ch17_markov_switching

Bitcoin: Two Regimes (1/2)



MFM_ch17_markov_switching

Bitcoin: Two Regimes (2/2)

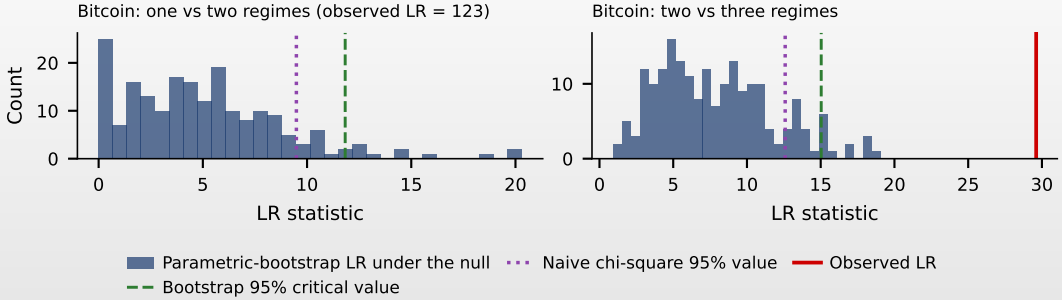
- Weekly log returns 2014–2026, 626 weeks
- Calm: $\hat{\mu} = 0.26\%$ per week, $\hat{\sigma} = 2.93\%$, duration 3.8 weeks
  - Turbulent: $\hat{\mu} = 1.20\%$ (SE 0.58), $\hat{\sigma} = 11.22\%$
  - Turbulent duration 6.3 weeks, 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean
  - Regimes are short: volatility states, not bubble phases
- Regimes ignore the price level: a slow, calm bubble stays calm
  - BSADF says "the price is accelerating"; the regime says "risk has changed"

MFM_ch17_markov_switching

How Many Regimes? A Bootstrap LR Test (1/3)

- H_0: one regime (iid, independent and identically distributed, Normal returns) vs H_1: two Markov regimes
  - Statistic $\mathrm{LR} = 2(\hat{\ell}_2 - \hat{\ell}_1)$; $\hat{\ell}_1$, $\hat{\ell}_2$: maximised log-likelihoods with one and two regimes
  - A large LR favours two regimes
- Non-standard null distribution (Appendix) Hansen (1992)
  - p_{11}, p_{22} are not identified under H_0, so LR is not χ^2
- **Parametric bootstrap**
  - Simulate from the fitted H_0 model, refit both models, recompute LR; $B = 199$ replications
  - $p = (1 + \#\{\mathrm{LR}^* \geq \mathrm{LR}\})/(B + 1)$; $\#\{\cdot\}$: number of bootstrap values LR^* at least as large as the observed LR

How Many Regimes? A Bootstrap LR Test (2/3)



MFM_ch17_inference

How Many Regimes? A Bootstrap LR Test (3/3)

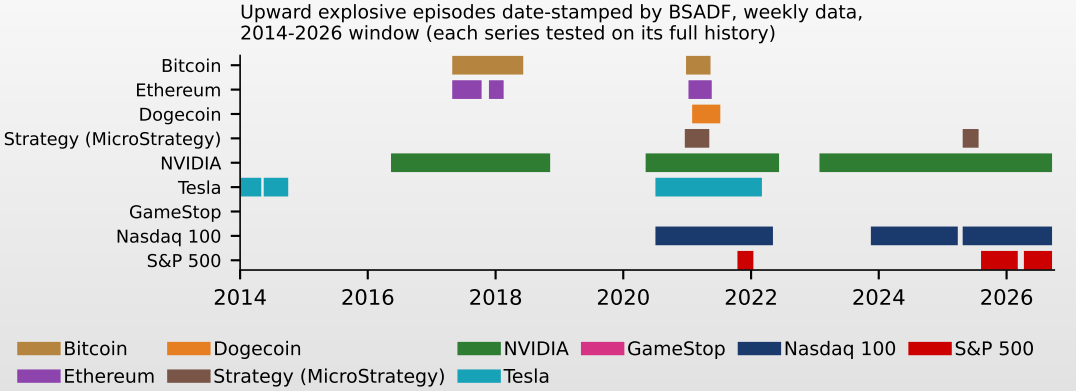
- One vs two regimes
  - Bitcoin: LR $= 123.2$ (95% value 11.8, $p = 0.005$); Nasdaq 100: LR $= 475.7$ (11.1, $p = 0.005$)
- Two vs three regimes, Bitcoin (replication target of Seminar 17, B6)
  - LR $= 29.6$; bootstrap 95% value 15.0 vs naive χ^2_6 value 12.6; $p = 0.005$
- One regime is rejected for both series; a third Bitcoin regime is also supported
  - Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes

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Are Crypto-Assets Bubbles?

- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero Cheah & Fry (2015)
- Alternative: value from network adoption (Metcalfe's law), with LPPL bubbles on top Pele & Mazurencu-Marinescu-Pele (2019)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble Pele et al. (2023)
- Without a fundamental value we can still test the price: is it explosive, and when?

Explosive Episodes in Crypto and Tech Stocks (1/2)



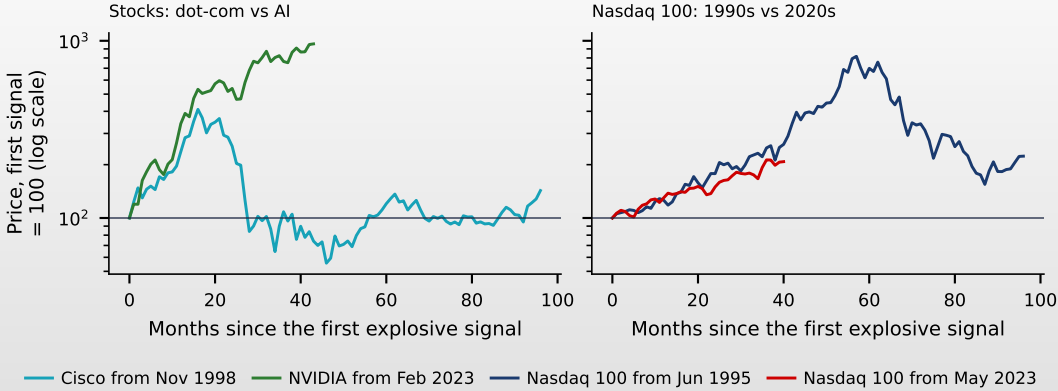
MFM_ch17_crypto_ai

Explosive Episodes in Crypto and Tech Stocks (2/2)

- Weekly log closes; rising episodes only
  - Each series tested on its own history, with its own Monte Carlo critical values
- Crypto booms cluster in 2017 and late 2020–2021
  - Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021
- Stocks
  - Strategy (formerly MicroStrategy; holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
  - GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration
  - Indices: Nasdaq 100 Jul 2020–May 2022; Nov 2023–Mar 2025; Apr 2025–ongoing; S&P 500 Oct 2021–Jan 2022; Aug 2025–Mar 2026; Apr 2026–ongoing
- An accelerating **fall** also gives $b > 0$: always report the direction

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Dot-Com vs AI (1/2)



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Dot-Com vs AI (2/2)

- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
  - First signal: start of the first date-stamped run beginning in 1995 or later (dot-com) or in 2023 or later (AI), not of the run that contains the peak
- Cisco: first signal Nov 1998; NVIDIA: first signal Feb 2023
  - Cisco peaked after 16 months at 4.1 times the signal price
  - NVIDIA, 43 months later: 9.6 times, episode still open
- Nasdaq 100: the AI rise is slower than in the 1990s
  - From Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months

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Is There an AI Bubble? What the Tests Can Say

- Monthly BSADF flags NVIDIA and the Nasdaq 100
  - NVIDIA: Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing; Nasdaq 100: current episode from May 2023
  - Monthly dates ($T = 332$, $w_0 = 36$, $L = 6$ months); the weekly timeline ($T = 1\,443$, $w_0 = 82$, $L = 8$ weeks) shifts the dates by up to a few months, because T, w_0 and L change with the frequency
- The tests cannot separate a bubble from explosive fundamentals Pástor & Veronesi (2009)
  - Fast-growing expected profits of a new technology can also make prices explosive
- History
  - Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- What we can say
  - The tests detect explosive price dynamics, not their cause
  - The industry run-up evidence Greenwood, Shleifer & You (2019) motivates watching crash risk, but does not quantify it for these stocks
  - The timing of any crash is unknown

Case Study (1/8): the Research Question

- Paper: DeFusco, Nathanson & Zwick (2022), *Journal of Financial Economics*, 146(1), 205–229
  - Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Questions (Introduction)
  - Who buys during a boom?
  - Can speculation explain the joint path of prices, transaction volume and unsold listings?
- Speculation, in the Harrison–Kreps definition the authors adopt (footnote 1)
  - Paying more for an asset because one has the right to resell it
- Three phases (Section 2 of the paper)
  - Boom: prices and volume rise
  - "Quiet": volume falls while prices still rise and unsold listings pile up
  - Bust

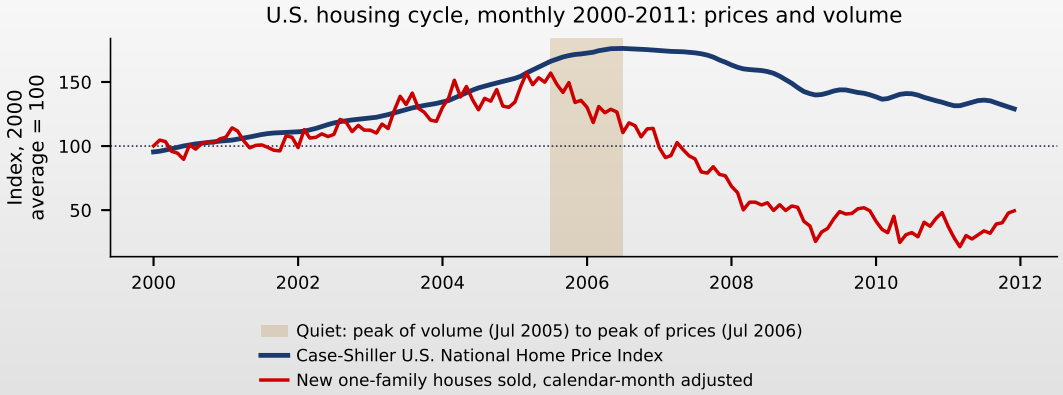
Case Study (2/8): Why It Matters for This Chapter

- The explosiveness tests of this chapter use prices only
  - Bubbles for Fama also study turnover, issuance and volatility Greenwood, Shleifer & You (2019)
- Bubbles come with high turnover Scheinkman & Xiong (2003)
  - They trade the resale option of Bubbles Beyond Rational Expectations
- Quantities show who drives a boom
  - Blockchain flows did so for the 2017 Bitcoin boom Griffin & Shams (2020)
- This paper adds the housing evidence
  - Who the speculators are, when they trade and how much of the cycle they explain

Case Study (3/8): Data, Method and Headline Results

- Data (Section 1.1): 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas
  - **Speculators** (Section 1.2): short-term buyers who resell within three years, and non-occupant buyers
- Lead–lag (Eq. 1): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$, $k = -12, \ldots, 48$ months
  - i: metropolitan area; t: month; p: demeaned log price; v: demeaned sales rate; $\eta_{i,t}$: error
  - $k > 0$: volume k months earlier; implied correlation $\beta_k \, \mathrm{sd}(v_{i,t-k})/\mathrm{sd}(p_{i,t})$, sd: standard deviation
- Figures 2 and 4
  - The correlation peaks at a lag of 24 months
  - Short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: cities with more speculation had larger booms and larger busts
  - One standard deviation more speculative volume: boom larger by 24.9 points, bust deeper by 7.4 points
- Model (Sections 5–6, Table 6): extrapolating short-horizon buyers amplify the cycle
  - A 5% purchase tax cuts the bust from 8.2% to 1.1%

Case Study (4/8): Prices and Volume, 2000–2011



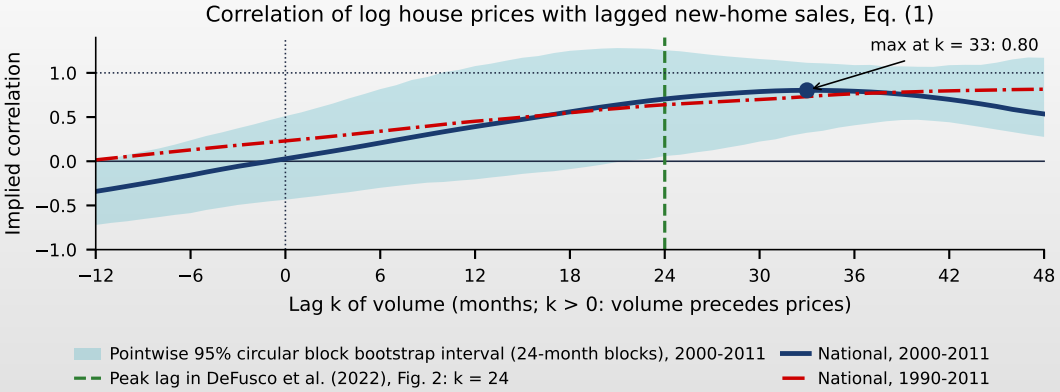
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Case Study (5/8): Prices and Volume, 2000–2011

- FRED (Federal Reserve Economic Data, St. Louis Fed): Case-Shiller U.S. National Home Price Index and new one-family houses sold
  - 2000 average = 100; volume adjusted by calendar-month means, as in Figure 1 of DeFusco, Nathanson & Zwick (2022)
- Boom: sales peak at 157 in Jul 2005
  - Paper: 150% of the 2000 level; prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak
  - Sales already down to 110 at the price peak (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices
  - New-home sales exclude resales, so they cannot show who trades

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Case Study (6/8): Volume Leads Prices



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Case Study (7/8): Volume Leads Prices

- Eq. (1) on the national series, without a constant
  - Pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- 2000–2011: maximum at $k = 33$ (0.80, CI $[0.32, 1.11]$)
  - At the paper's $k = 24$: 0.70, CI $[0.06, 1.26]$
- The band excludes zero for $k = 22, \ldots, 48$
  - But one national cycle holds few independent blocks: wide intervals
- Direction as in the paper: volume first, by two to three years
  - The length is not precise (1990–2011: $k = 48$)

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Case Study (8/8): Limits and Open Questions

- Data limit of our illustration (Seminar 17, C3)
  - The public series are national and cover new homes only: short-term and non-occupant volume cannot be built
  - Figures 1–2 can be illustrated, Table 2 cannot
- Limits stated by the authors
  - Holding periods are realised, not intended (footnote 4; instrumental-variable check in the Online Appendix)
  - The non-occupant proxy misclassifies owners who give the property address for tax mail (Section 1.2)
  - Selling horizons are exogenous in the model, so the counterfactuals may overstate short-term buyers (Section 4.3)
  - Credit and new construction are left out (Sections 6.5.3, 7)
- Open questions
  - Does trading volume also lead prices in crypto booms?
  - Is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

Can Bubble Alarms Warn of Crashes?

Real-time bubble alarms

- Questions
  - Can bubble alarms be raised in real time with a controlled false-alarm rate?
  - Do they carry crash information?
- BSADF date-stamping: one right-tailed test per date
  - Size controlled date by date only (Size of GSADF When Volatility Changes: Results)
  - Live research: monitoring rules for bubbles *and* crashes Whitehouse, Harvey & Leybourne (2023); "econometric thermometers" for housing Shi & Phillips (2023)
- Formal version: alarm $A_t \in \{0, 1\}$, crash $C_{t,h}$ = a fall of at least 20% within $h = 90$ days
  - H_0: $P(C_{t,h} \mid A_t = 1) = P(C_{t,h} \mid A_t = 0)$; H_1: larger after an alarm
  - Falsified if the hit rate after alarms does not exceed the rate without alarms, with inference robust to overlapping windows
- Identification: explosive prices need not be a bubble
  - Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009)
  - Run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

The Discovery Loop and Its Checks

- **Literature**
  - Ask for papers on the false-alarm rate of sequential bubble monitoring
  - Resolve every DOI (Digital Object Identifier, the permanent identifier of a paper) on Crossref; find the table the answer quotes
- **Hypotheses**, each with its own observable implication
  - Rational bubble, explosive expected cash flows, falling discount rate
  - E.g. an explosive price-to-earnings ratio vs explosive earnings
- **Code**
  - Replicate a known number first (GSADF $= 3.85$ for weekly Bitcoin), then extend
- **Robustness** and **critique**
  - Pre-register the grid of horizons and crash thresholds, with a Holm correction
  - Holm: the k-th smallest of m p-values is compared with $0.05/(m-k+1)$, stopping at the first non-rejection
  - Ask the assistant to attack the result as a referee
- **Checks** the researcher must run (details and prompts in the Appendix)
  - No full-sample information in a date stamp; right-tail critical values; wild bootstrap under changing volatility
  - Multiplicity over dates and assets; inference robust to overlapping outcome windows

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal
  - "A positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher's exact test"
  - Alarms come in runs and outcomes overlap, so we test by circular rotations of the alarm series
- Hit rates after alarms and without: Bitcoin: Real-Time Confidence Indicator
  - Bitcoin: difference -15.5 percentage points; one-sided $p = 0.96$ for $H_1 : \Delta > 0$ (larger after an alarm) from 658 rotations
  - Nasdaq 100: difference -7.1 points, $p = 0.98$
- Pre-registered grid of 4 horizons $\times$ 3 thresholds
  - Smallest raw p: 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
  - No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation (assumptions in the Appendix)
  - The indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal

Your Turn: a Project Seed

- **Questions**
  - Can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period?
  - Can it still warn before large falls?
- **Replicate first**
  - GSADF = 3.85 for weekly Bitcoin
  - The crash probability after run-ups above 100% in Greenwood, Shleifer & You (2019) (ours: 46%)
- **Extend**
  - Calibrate the alarm threshold by simulation over the whole monitoring period Whitehouse, Harvey & Leybourne (2023)
  - Evaluate the alarms with rotations or a block bootstrap
- **Deliverables**
  - A pre-registered grid, fixed in a dated document before the evaluation
  - A notebook that regenerates every number; false-alarm and hit rates with valid p-values
- **AI log** (templates)
  - `AI_USE.md`: the prompts that shaped the design
  - `AI_ERRORS.md`: each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it

Key Takeaways

- A bubble is price minus fundamental value
  - Rational bubbles must be explosive; bursting bubbles hide from full-sample tests
- GSADF tests for explosive episodes; BSADF date-stamps them in real time
  - Late volatility rise: Monte Carlo critical values reject 32.0% of true nulls, the wild bootstrap 4.1%
  - Date-stamping over many dates needs family-wise control
- Across our episodes the tests warned months before the peak, but did not time the crash
- LPPLS describes the acceleration
  - Its critical time moves with the window
  - The confidence indicator is judged by a dependence-robust test of hit rates
- Sharp run-ups raise crash risk without a statistically established average loss
  - A risk-management signal, not a trading rule

Which Tool When?

Question	Tool
Has there been any explosive episode?	GSADF with simulated critical values
When did it start and end?	BSADF date-stamping, minimum duration $\lceil \ln T \rceil$; family-wise critical values for long monitoring
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets (Greenwood, Shleifer and You 2019)

Check Yourself, and Next: Chapter 18

Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically? revisit
- Why is the 95% critical value of GSADF far above that of a single ADF? revisit
- Why is a single LPPLS critical time not a crash forecast? revisit
- Why must hit rates after alarms be tested with a block bootstrap or rotations? revisit

Chapter 18: systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- How much one institution adds to the risk of the system
  - CoVaR: the value-at-risk (VaR) of the system when an institution is in distress
  - SRISK: the capital shortfall of an institution in a crisis
- Further reading: Franke, Härdle & Hafner (2019), Kindleberger & Aliber (2005), Shiller (2015), Sornette (2017)

Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself h times: $P_t = \sum_{i=\mathbf{1}}^{h} \frac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \frac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$ (law of iterated expectations)
- Let $h \to \infty$: $P_t = F_t + B_t$ with $B_t = \lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition**: $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$; it holds $\Leftrightarrow$ $B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period: $\mathbb{E}_t[B_{t+\mathbf{1}}] = \lim_h \mathbb{E}_t[P_{t+\mathbf{1}+h}]/(1+r)^h = (1+r)\,B_t$, the condition that the discounted bubble is a martingale
- Seminar 17, Exercise A1 derives the general solution and why $B_t \geq 0$ rules out a bubble that starts from zero Diba & Grossman (1988)

Back

Bubbles for Fama: Effective Sample Size

- The run-ups fall in 31 calendar years and overlap: consecutive 2-year windows of the same boom share months
- Year-block bootstrap SE (9.1 percentage points) vs binomial SE (6.9): a design effect (squared ratio of the two SEs) of 1.7, i.e. about 30 independent observations
- Blocks of 2, 3 and 5 consecutive years (2-year windows of adjacent years overlap): SE $= 8.9, 8.7, 8.3$
- Unconditional mean 2-year return over all industry-months: 28%
  - Above the CI of the mean return after a 100% run-up (-14% to 20%): returns after run-ups look lower than usual
  - A descriptive comparison, not a formal test of the difference of two dependent estimates

Back

Why the Full-Sample ADF Fails: the Leverage of Collapses

- OLS slope: $\hat{b} = \sum_t \tilde{y}_{t-\mathbf{1}}\, \Delta y_t \big/ \sum_t \tilde{y}^{\mathbf{2}}_{t-\mathbf{1}}$
  - $\tilde{y}_{t-\mathbf{1}}$: the lagged log price minus its sample mean
  - Expansion: $\Delta y_t > 0$ while $\tilde{y}_{t-\mathbf{1}}$ rises above its mean: positive terms
  - Collapse: the most negative Δy_t at the highest $\tilde{y}_{t-\mathbf{1}}$: a few very large negative terms
- Evans form in logs, $y = \ln B$, collapse to $\delta = e^c$ with probability $1 - \pi$ Evans (1991)
  - For $B \gg \delta$ the survival log growth is $\approx \ln \frac{\mathbf{1}+r}{\pi}$
  - $\mathbb{E}[\Delta y_t \mid y_{t-\mathbf{1}}] \approx \pi \ln \frac{\mathbf{1}+r}{\pi} + (1-\pi)(c - y_{t-\mathbf{1}})$
  - Slope $-(1-\pi) < 0$: apparent mean reversion, although $\mathbb{E}_t B_{t+\mathbf{1}} = (1+r)B_t$
- Windows with the expansion but not the collapse have $\hat{b} > 0$ when the log price accelerates (next slide)
  - A supremum over windows restores the power Phillips, Shi & Yu (2015b)

Back

Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**: $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$, an autoregressive root $1+r>1$
  - In logs a pure bubble path $B_t = B_0(1+r)^t$ has constant growth: $\ln B_t = \ln B_0 + t\ln(1+r)$, a deterministic linear trend in logs, $b=0$ in $\Delta y_t = a + b\,y_{t-1} + \varepsilon_t$
- The log price $\ln(F_t + B_t)$ **accelerates** while the bubble share grows: its log growth rises from that of F towards $\ln(1+r)$ (a convex log path)
  - Once the bubble dominates, log growth is roughly constant again: the log-price regression then has little power
- So the log-price test targets a different alternative, **accelerating** log prices, not the level condition itself; the specification of the null and of the deterministic terms matters Phillips, Shi & Yu (2014)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq); PSY test the price-dividend ratio Phillips, Wu & Yu (2011), Phillips, Shi & Yu (2015a)
- Numerical version on the main path: Levels Explode, Log Prices Accelerate

Back

Limit Theory (1): the Null Distribution

- Null model: $y_t = d\,T^{-\eta} + y_{t-1} + \varepsilon_t$, $\eta > 1/2$, ε_t iid $(0, \sigma^2)$ Phillips, Shi & Yu (2015a)
  - A random walk with an asymptotically negligible drift $d\,T^{-\eta}$ (d: a constant)
- Limit (PSY, Theorem 1, eq. 5), by the functional central limit theorem and continuous mapping
  - $$\mathrm{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w \left[W(r_2)^2 - W(r_1)^2 - r_w\right] - \int_{r_1}^{r_2} W\,dr\,\left[W(r_2) - W(r_1)\right]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W^2 dr - \left(\int_{r_1}^{r_2} W\,dr \right)^2 \right\}^{1/2}}$$
  - $r_w = r_2 - r_1$; W: standard Brownian motion; $\Rightarrow$: convergence in distribution as $T \to \infty$
- σ and d disappear: the limit is pivotal
  - SADF and GSADF are suprema of the same functional: their critical values depend only on r_0 and the deterministic terms
- $r_1 = 0$, $r_2 = 1$: the Dickey–Fuller limit with a constant
  - Its 95% quantile is negative (simulated: -0.06), far from 1.645
  - The drift and intercept specification changes the limit and the size Phillips, Shi & Yu (2014)

Back

Limit Theory (2): Mildly Explosive Alternatives

- Alternative: $y_t = \rho_T\, y_{t-1} + \varepsilon_t$ with $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \to \infty$, $k_T = o(T)$; PSY use $k_T = T^{\alpha}$, $\alpha \in (0, 1)$
  - c: how explosive; k_T: the speed at which the root approaches 1; $k_T = o(T)$: $k_T / T \to 0$
  - Closer to 1 than any fixed explosive root, farther than a local-to-unity root $1 + c/T$: the "mild" speed of real bubbles
- Phillips and Magdalinos: $\dfrac{\rho_T^T}{\rho_T^2 - 1}\,(\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$, a standard Cauchy law, without Gaussian errors Phillips & Magdalinos (2007)
  - $\hat{\rho}_T$: the OLS estimate of ρ_T; $\mathcal{C}$: the standard Cauchy distribution
  - Rate $k_T \rho_T^T$: exponentially fast; the ADF statistic diverges to $+\infty$
- Hence SADF and GSADF diverge too (for GSADF any window of at least $r_0 T$ observations inside the bubble suffices): the tests are consistent, power $\to 1$ Phillips, Shi & Yu (2015b)
- Finite samples: power depends on c, α and the duration; short bursts may stay below the bar, as in the Evans simulation

Back

Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8)
  - $\hat{r}_e = \inf\{r_2 \geq r_0 : \mathrm{BSADF}_{r_2} > scv_{r_2}^{\beta_T}\}$, $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \mathrm{BSADF}_{r_2} < scv_{r_2}^{\beta_T}\}$
  - $\hat{r}_e$, $\hat{r}_f$: estimated start and end, as fractions of the sample; inf: the first r_2 that meets the condition
  - $scv_{r_2}^{\beta_T}$: the 100$(1-\beta_T)$% quantile of BSADF on $\lfloor Tr_2 \rfloor$ observations; β_T: the level; $\lfloor \cdot \rfloor$: rounded down
- Consistency (PSY, eq. 11): $(\hat{r}_e, \hat{r}_f) \to_p (r_e, r_f)$ if $\dfrac{1}{scv^{\beta_T}} + \dfrac{scv^{\beta_T}}{T^{1-\alpha/2}} \to 0$
  - $\to_p$: convergence in probability to the true dates r_e, r_f
  - The critical value must diverge, so that false alarms vanish ($\beta_T \to 0$)
  - But slower than $T^{1-\alpha/2}$, so that true bubbles are still found; e.g. like $\ln T$ Phillips, Shi & Yu (2015a)
- In practice β_T is fixed at 5% at every date
  - Over hundreds of dates a false alarm becomes likely (the Monte Carlo size study in the wild bootstrap section)

Back

The Price-Dividend Ratio: Comparison with PSY

- Our critical values: 1 000 Monte Carlo paths of the sample length
- PSY, on data to 2010 and without our filter of 8 consecutive months, also date 1879–1880, 1955–1956 and 1986–1987 Phillips, Shi & Yu (2015a)
- With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too (Seminar 17, B1)

Back

Family-Wise Critical Values: Bootstrap Design

- Phillips and Shi: the 95% quantile of the maximum of BSADF over τ_b consecutive dates Phillips & Shi (2020)
  - Sets the FWER at 5% for the bootstrap design: τ_b monitoring dates after a start-up of w_0 observations
- Resampling
  - Residuals e_t of $\Delta y_t = a + e_t$, drawn with replacement and multiplied by $N(0,1)$ weights
  - Paths of $w_0 + \tau_b - 1$ changes; 999 replications
- Why the control is only approximate here
  - BSADF over the full history: later dates search more start points than the bootstrap design
  - Drawing with replacement loses the time order: unlike the pointwise wild bootstrap, which keeps each Δy_t at its date, these paths ignore the volatility clustering of Bitcoin
- The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

Back

Real Time vs Hindsight: Caveats

- Column definitions as on the main slide, with three caveats
  - Confirmed is only pseudo real time: w_0 and L come from the full-sample T (a truly real-time design fixes them ex ante)
  - GameStop: leads in calendar days converted to months
  - GameStop: fall to the lowest close within 1 year, not 2
- Bitcoin 2021 and the two Nasdaq 100 falls: explained on the main slide Reading the Real-Time Table

Back

Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate Astill et al. (2018)
  - Astill et al. compare monitoring statistics with the maximum over a training period
  - The false-positive rate is then set for a chosen horizon, robust to time-varying volatility
- Bubbles and crashes Whitehouse, Harvey & Leybourne (2023)
  - After a bubble is detected, monitor for the crash with the sign of the mean of Δy_t
  - It never signals a crash before the crash starts
- Earlier detection Whitehouse, Harvey & Leybourne (2025)
  - Alternative variance standardisations keep the false-detection probability but find bubbles sooner (Bitcoin, housing)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery Phillips & Shi (2018)
- Alternatives to sup-ADF
  - A Chow-type break test, random walk to explosive, has the highest power in Homm & Breitung (2012)
  - Recursive tests with a start-point search Harvey, Leybourne & Sollis (2015)

Back

LPPL in the Cryptocurrency and Romanian Markets

- Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash Pele, Mazurencu & Nijkamp (2013); crash modelling for the BVB Pele & Mazurencu-Marinescu (2012)
- Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles Pele & Mazurencu-Marinescu-Pele (2019), Conda et al. (2026)
- Dogecoin 2021: LPPL together with sentiment from social media Pele (2025)
- Common lesson: LPPL describes the acceleration well; the crash date remains the weak point

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Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood of θ (all μ_j, σ_j, p_{ij}), by prediction and update Hamilton (1989)
  - $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1})\, \varphi(r_t; \mu_j, \sigma_j^2)$
  - $\mathcal{F}_t$: returns up to t; $\varphi(x; \mu, \sigma^2)$: the Normal density with mean μ and variance σ^2
- Kim smoother, backwards Kim (1994)
  - $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \dfrac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$
- EM algorithm: the M-step is a weighted average Hamilton (1990)
  - Smoothed $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$, $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$
  - $\hat{\mu}_j = \dfrac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}$, $\hat{\sigma}_j^2 = \dfrac{\sum_t \xi_{tj}(r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}$, $\hat{p}_{ij} = \dfrac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- Unbounded likelihood ($\sigma_j \to 0$ on one return) and several local maxima
  - Best interior maximum from 20 random starts; no $\hat{\sigma}_j$ may collapse towards 0, also in bootstrap refits
  - Standard errors from the inverse numerical Hessian assume a known number of regimes

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Why the LR Test for Regimes Is Non-Standard

- Under H_0 the regime means and variances coincide Hansen (1992)
  - p_{11}, p_{22} are not identified: nuisance parameters present only under H_1
  - The information matrix is singular and some scores are identically zero
- Hansen: bounds on the standardised LR via empirical-process theory Hansen (1992)
- Carrasco, Hu and Ploberger: an optimal test Carrasco, Hu & Ploberger (2014)
  - It needs only the estimate under H_0, with bootstrap critical values
- Two vs three regimes: the same logic
  - The null model is then the fitted two-regime chain

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Case Study, Data and Method

- Data (Section 1.1): arm's-length sales only
  - 115 metropolitan areas (48% of the US housing stock); listings linked to sales in 57 of them
  - The cycle analysis covers 2000–2011: realised holding periods need consecutive sales
- Speculators (Section 1.2)
  - Short-term buyers: resell within three years (realised holding period)
  - Non-occupant buyers: a mailing address different from the property address (102 areas)
- Lead–lag (Eq. 1, Figure 2): construction of the variables
  - p: log price, demeaned by city; v: sales over the 2000 housing stock, demeaned by city and calendar month
  - Standard errors clustered by month
- Cross-section (Table 2): OLS across cities on the speculative volume boom (2000–2005)
  - Outcomes: price boom (2000–2006), price bust (2006–2011), listings, volume, foreclosures
- Model (Sections 5–6): extrapolating buyers with different horizons; not every listing sells
  - Counterfactuals: rational expectations, no speculators, transaction taxes

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Case Study, Complete Results (1/2)

- Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005)
  - Short-term sales: from 510 to 940 thousand, 39% of the rise; non-occupant purchases: 53%
  - Interpretation: speculators account for much of the volume boom
- Table 2, Panel A: price-boom coefficient 1.930 (standard error 0.297, $R^2 = 0.272$)
  - One standard deviation more (12.9%, Table 1): price rise larger by 24.9 percentage points
  - Price bust -0.571 (0.083): 7.4 points deeper
  - Interpretation: more speculation, larger booms and larger busts

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Case Study, Complete Results (2/2)

- Table 2, Panels B–C (quantities in % of 2003 total volume; Section 3.2)
  - Unsold listings in the quiet: 5.961 (1.353), 76.9 points more per standard deviation
  - Foreclosures: 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas
  - Interpretation: speculators who bought in the boom list in the quiet and fail to sell
  - That this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year goes with an 8.2 points larger short-term buyer share
  - Interpretation: extrapolation draws short-horizon buyers in
- Table 6 (model): a 5% purchase tax
  - On all buyers: the price bust falls from 8.2% to 1.1%; on non-occupants only: to 5.8%
  - Interpretation: short horizons, not non-occupancy, amplify the cycle

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Case Study, Lead–Lag Inference

- p and v demeaned over the full sample; $k = -12, \ldots, 48$ months
  - Implied correlation $\hat{\beta}_k \, \mathrm{sd}(v_{t-k})/\mathrm{sd}(p_t)$
- At $k = 0$: 0.03; at $k = -12$: -0.34
  - Level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean
  - The "implied correlation" is a standardised slope, not a Pearson correlation: bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series
  - Doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory
- 1990–2011: the maximum moves to the edge of the grid, $k = 48$ (0.82)

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The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: map what is known, then verify every reference
  - "Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design."
  - Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses**: competing explanations, each with its own observable
  - "For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts."
  - E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code**: on the course data, replicating a known number first
  - "Implement GSADF with r0 = 0.01 + 1.8/sqrt(T) on weekly Bitcoin closes from the course data and simulate its 95% critical value."
  - Replicate first: GSADF = 3.85 and critical value 2.27 on 626 weeks, as in this chapter

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
  - "`Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.`"
  - Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
  - "`Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?`"
  - Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid *p*-values, all 12 cells, the alarms that failed
  - The prompts that shaped the design go into `AI_USE.md`

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What the Researcher Must Check

- **Full-sample information in date-stamping**
  - Critical values, lag order or detrending fitted on the whole sample
  - Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail**
  - Left-tail ADF critical values (about -2.86) used against an explosive alternative
  - Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility**
  - Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity**: many dates, assets and windows
  - Test: run the rule on simulated random walks and count paths with at least one alarm
- **Overlapping outcomes**: Fisher's exact or χ^2 tests overstate significance
  - Test: block bootstrap or circular rotations of the alarm series; drop windows cut by the end of the data

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Mini-Case: What the Result Does and Does Not Say

- A large one-sided p is not evidence of **no** information
  - It only fails to support a **higher** crash probability after alarms
  - Both point estimates are **negative**: one-sided rotation p for $\Delta < 0$, a lower crash rate after alarms, 0.07 (Bitcoin), 0.03 (Nasdaq 100)
  - Bitcoin: this 0.07 and the Newey–West one-sided 0.024 answer the same question under different null assumptions (stationary rotations vs a regression with HAC (Heteroskedasticity and Autocorrelation Consistent) errors)
  - Post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days
- Assumption behind the rotation p-values
  - Under H_0 every circular shift of the alarm series (at least 26 steps, ends joined) is as likely as the observed alignment
  - Stronger than equal hit rates: it needs stationary series and alarms independent of the crash process
  - Hence exploratory p-values; check their size by simulation under plausible null processes
- Block-bootstrap CI and Newey–West test of the same hit rates: Seminar 17, B7

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Further Study on Quantinar

Advanced courses on quantinar.com

- Cryptocurrency: Speculative Asset and Medium of Exchange
- Are Cryptos becoming alternative Assets?
- Statistics of Financial Markets

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **BET**: Bucharest Exchange Trading (index)
- **BET-FI**: BET Financial Investment companies index
- **BSADF**: Backward Sup Augmented Dickey–Fuller (statistic)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **EM**: Expectation-Maximisation (algorithm)
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **JLS**: Johansen–Ledoit–Sornette (model)
- **LPPL**: Log-Periodic Power Law
- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **LR**: Likelihood Ratio (test)
- **MC**: Monte Carlo (simulation)
- **NVIDIA**: NVIDIA Corporation (company name, US chip designer)
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- **PWY**: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- **SADF**: Sup Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SRISK**: Systemic RISK (capital shortfall measure)
- **US**: United States

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